

The torus as a quotient space

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In this article I will give a sketch of a proof that we can construct the torus $T^2 = S^1 \times S^1$ as the quotient of the closed unit square $[0, 1] \times [0, 1]$.

For this proof we will need the following:

Theorem. Any continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Proof. Let X be compact, Y Hausdorff and $f: X \rightarrow Y$ a continuous bijection. We need to show that f^{-1} is continuous. This is equivalent to showing that $f(U)$ is open for every open subset $U \subseteq X$. This is equivalent to showing that $f(U)$ is closed for every closed subset $U \subseteq X$. Let $U \subseteq X$ be closed. Then U is a closed subset of a compact space, so is compact. Hence $f(U)$ is compact since it is the continuous image of a compact set. Then $f(U)$ is a compact subspace of a Hausdorff space, so is closed. This completes the proof. $\square$

Also, to make the argument go through we need the Heine-Borel theorem, which I won't prove here.

Theorem (Heine-Borel). Every closed bounded subset of $\mathbb{R}^n$ is compact.

Finally, we need the universal property of quotients.

Theorem (Universal property of quotients). Let $\sim$ be an equivalence relation on a space X and let q be the associated quotient map. Let $f: X \rightarrow Y$ be a continuous map which is constant on the fibers of the quotient. Then, there is a unique continuous map $\hat{f}: \frac{X}{\sim} \rightarrow Y$ such that $f = \hat{f} \circ q$.

Now the argument will work. Let S^1 be the circle $\{(x, y) \mid x^2 + y^2 = 1\} \subseteq \mathbb{R}^2$ with the subspace topology from $\mathbb{R}^2$. Let $T^2 = S^1 \times S^1$ in the product topology. Finally, let $I = [0, 1]$ be the closed unit interval with the usual subspace topology on $\mathbb{R}$. Consider the equivalence relation $\sim$ on $I^2 = [0, 1] \times [0, 1]$ given by

- $(0, s) \sim (1, s)$ for all $s \in [0, 1]$
- $(t, 0) \sim (t, 1)$ for all $t \in [0, 1]$.

By this I mean the smallest equivalence relation on I^2 such that the above two conditions hold. I claim that

$$\frac{I^2}{\sim} \cong T^2.$$

To show this, first define the function $f: I \rightarrow S^1$ by $f(t) = (\cos(2\pi t), \sin(2\pi t))$. This is continuous because both $\cos$ and $\sin$ are continuous. It also surjects onto S^1 . Define $g: I^2 \rightarrow T^2$ by $g(x, y) = (f(x), f(y))$. This is a continuous surjection onto T^2 because it is a continuous surjection in each co-ordinate by preceding argument. Let us inspect how g interacts with the quotient $\sim$. The equivalence classes of $\sim$ are

- $[(x, y)] = \{(x, y)\}$ when $x \in (0, 1)$ and $y \in (0, 1)$.
- $[(0, y)] = [(1, y)] = \{(0, y), (1, y)\}$ when $y \in (0, 1)$.

- $[(x, 0)] = [(x, 1)] = \{(x, 0), (x, 1)\}$ when $x \in (0, 1)$.
- $[(0, 0)] = [(0, 1)] = [(1, 0)] = [(1, 1)] = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$.

We can see that g is constant on the fibers of this quotient, since $\cos(0) = \cos(2\pi)$ and $\sin(0) = \sin(2\pi)$. Therefore, by the universal property of quotients, there exists a continuous map $\hat{g}$ such that $g = \hat{g} \circ q$, where q is the quotient map associated with $\sim$. The induced map $\hat{g}$ must be surjective because g is surjective. In addition, $\hat{g}$ must be injective, because g maps two points of I^2 to the same point of T^2 if and only if the points are $\sim$ -equivalent. It follows that $\hat{g}$ is a continuous bijection $\frac{I^2}{\sim} \rightarrow T^2$.

Observe that T^2 is a Hausdorff space, because each copy of S^1 is Hausdorff, being a subspace of $\mathbb{R}^2$. Furthermore, I^2 is compact by the Heine-Borel theorem, being a closed bounded subspace of $\mathbb{R}^2$. So the quotient space $\frac{I^2}{\sim}$ is compact, being the continuous image of a compact space. Putting these together, the map $\hat{g}$ is a continuous bijection from a compact space to a Hausdorff space. This implies $\hat{g}$ is a homeomorphism, establishing the result.